

DOCKER – TASK – 01

- 1) Setup a virtual machine using Ubuntu image.

The screenshot displays the AWS Management Console interface for launching a new EC2 instance. The 'Name and tags' section has 'docker-server' entered. The 'Application and OS Images (Amazon Machine Image)' section shows the 'Quick Start' tab with various OS options, including Ubuntu. The 'Summary' section on the right provides details: 1 instance, Canonical Ubuntu 26.04 AMI, m7i-flex.large instance type, default security group, and 1 volume of 20 GiB. A terminal window at the bottom shows the Ubuntu login prompt and system information as of Mon Aug 24 13:37:53 UTC 2026.

```
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Mon Aug 24 13:37:53 UTC 2026

System load:  0.08           Temperature:   -273.1 C
Usage of /:   11.1% of 10.25GB Processes:      158
Memory usage: 3%            Users logged in: 0
Swap usage:   0%            IPv4 address for enp3s0: 172.31.4.171

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

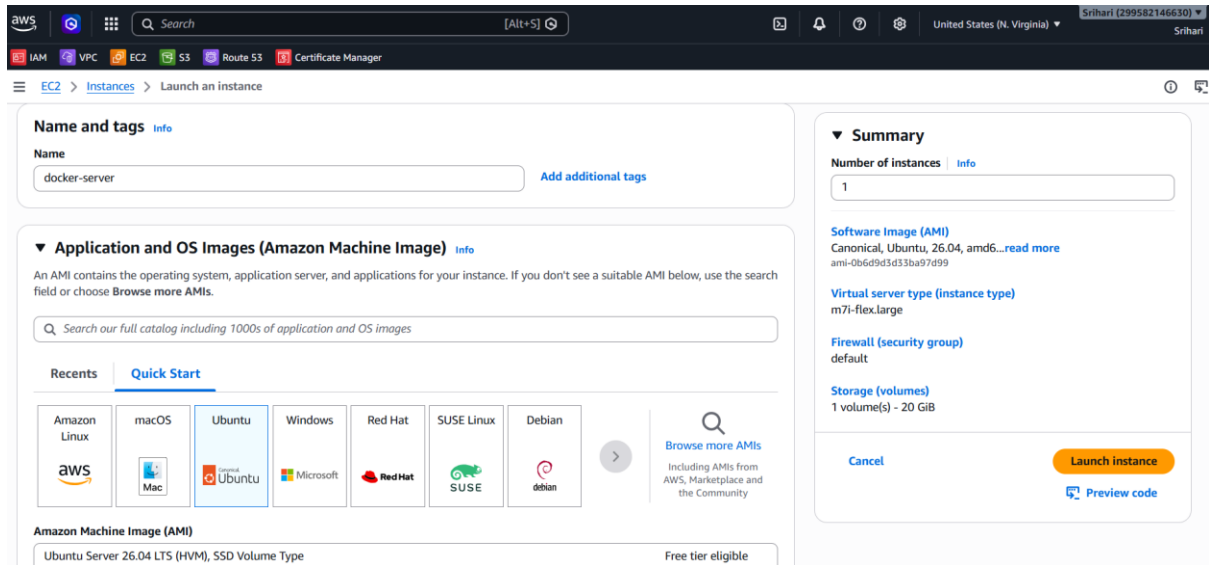
The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Mon Aug 24 13:35:43 2026 from 18.206.107.27
ubuntu@ip-172-31-4-171:~$
```

i-093103cc9c30873c1 (docker-server)
PublicIPs: 34.205.15.12 PrivateIPs: 172.31.4.171

- 2) Install docker on ec2.

AWS Console → EC2 → Launch Instance → type-large



Installed docker using

`sudo apt install docker.io -y`

```
root@ip-172-31-4-171:~# sudo apt install docker.io -y
Installing:
  docker.io

Installing dependencies:
  bridge-utils  containerd  dns-root-data  dnsmasq-base  pigz  runc  ubuntu-fan

Suggested packages:
  ifupdown  aufs-tools  cgroupfs-mount | cgroup-lite  debbootstrap  docker-buildx  docker-compose-v2  docker-doc  rinse  rootlesskit  zfs-fuse | zfsutils

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 115
  Download size: 74.0 MB
  Space needed: 281 MB / 17.2 GB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 pigz amd64 2.8-1build1 [68.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 bridge-utils amd64 1.7.1-4ubuntu3 [34.8 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 runc amd64 1.4.0-0ubuntu1 [9829 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 containerd amd64 2.2.2-0ubuntu1.1 [28.1 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 dns-root-data all 2025080400build1 [6022 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 dnsmasq-base amd64 2.92-lubuntu0.4 [441 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 docker.io amd64 29.1.3-0ubuntu4.1 [35.5 MB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 ubuntu-fan all 0.12.17 [34.3 kB]
Fetched 74.0 MB in 0s (153 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 85303 files and directories currently installed.)
Preparing to unpack .../0-pigz_2.8-1build1_amd64.deb ...
Unpacking pigz (2.8-1build1) ...
Selecting previously unselected package bridge-utils.
i-093103cc9c30873c1 (docker-server)
PublicIPs: 34.205.15.12  PrivateIPs: 172.31.4.171
```

Started docker and checked status by using systemctl commands

```
root@ip-172-31-4-171:~# docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
root@ip-172-31-4-171:~# sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-24 13:46:02 UTC; 3min 41s ago
   Invocation: ddc2fa2e36674a15b78b9e7e16c8b92b
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
   Main PID: 2781 (dockerd)
     Tasks: 9
    Memory: 25M (peak: 25.4M)
      CPU: 204ms
   CGroup: /system.slice/docker.service
           └─2781 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Aug 24 13:46:02 ip-172-31-4-171 dockerd[2781]: time="2026-08-24T13:46:02.009948229Z" level=info msg="Restoring containers: start."
Aug 24 13:46:02 ip-172-31-4-171 dockerd[2781]: time="2026-08-24T13:46:02.032142034Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Aug 24 13:46:02 ip-172-31-4-171 dockerd[2781]: time="2026-08-24T13:46:02.036607102Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Aug 24 13:46:02 ip-172-31-4-171 dockerd[2781]: time="2026-08-24T13:46:02.215596177Z" level=info msg="Loading containers: done."
Aug 24 13:46:02 ip-172-31-4-171 dockerd[2781]: time="2026-08-24T13:46:02.220005238Z" level=info msg="Docker daemon" commit=29.1.3-0ubuntu4.1 containerd-snapshotter
Aug 24 13:46:02 ip-172-31-4-171 dockerd[2781]: time="2026-08-24T13:46:02.220107477Z" level=info msg="Initializing buildkit"
Aug 24 13:46:02 ip-172-31-4-171 dockerd[2781]: time="2026-08-24T13:46:02.246920892Z" level=info msg="Completed buildkit initialization"
Aug 24 13:46:02 ip-172-31-4-171 dockerd[2781]: time="2026-08-24T13:46:02.249790325Z" level=info msg="Daemon has completed initialization"
Aug 24 13:46:02 ip-172-31-4-171 dockerd[2781]: time="2026-08-24T13:46:02.249843299Z" level=info msg="API listen on /run/docker.sock"
Aug 24 13:46:02 ip-172-31-4-171 systemd[1]: Started docker.service - Docker Application Container Engine.
root@ip-172-31-4-171:~#
i-093103cc9c30873c1 (docker-server)
PublicIPs: 34.205.15.12  PrivateIPs: 172.31.4.171
```

3) Pull 5 docker images .(nginx,apache tomcat,ubuntu,jenkins,sonarqube)

Pulled nginx by using: docker pull nginx command

```
root@ip-172-31-4-171:~# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
7eb55399d6de: Pull complete
746b934a8960: Pull complete
5d480233f531: Pull complete
f530c3e421fc: Pull complete
128fcc7b23b0: Pull complete
5508f6432d3e: Pull complete
26c307b5e35a: Pull complete
3976f7b8a9d7: Download complete
81dd0279e705: Download complete
Digest: sha256:0d4374c710a9649200e84f8ef8dbdd4fa76c0c107839cd50f1e42a63916b0f2e
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
root@ip-172-31-4-171:~#
```

docker pull tomcat

docker pull ubuntu

docker pull jenkins/jenkins

docker pull sonarqube

```
root@ip-172-31-4-171:~# docker pull tomcat
Using default tag: latest
latest: Pulling from library/tomcat
4f4fb700ef54: Pull complete
0926a8eb0e60: Pull complete
89416f64c8e9: Pull complete
b385a54fa58d: Pull complete
fbd4a7f21d48: Pull complete
6bd4a5f61a04: Pull complete
297337c1dc13: Pull complete
da76df643e43: Download complete
d477108507e1: Download complete
Digest: sha256:72400e3f4325d05f26a91bacf1ddce2237153ddb8a80304a493e749b6e292a89
Status: Downloaded newer image for tomcat:latest
docker.io/library/tomcat:latest
root@ip-172-31-4-171:~# docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
f3db1cd94078: Pull complete
06e9d71331fb: Pull complete
f79afc4bbefe: Download complete
Digest: sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
root@ip-172-31-4-171:~# docker pull jenkins/jenkins
Using default tag: latest
latest: Pulling from jenkins/jenkins
bffb3717e9ed: Pull complete
1da3cb2f93f2: Pull complete
3838de504c32: Pull complete
17ff5f3cef2e: Pull complete
7f7d90bd1824: Pull complete
d8a8dd6c1560: Pull complete
d856b8f48ad6: Pull complete
65cca0ec2bb2: Pull complete
d9173b50ffb3: Pull complete
a88f21496d6b: Pull complete
ec76481b7c7d: Pull complete
38e32ec4e36f: Pull complete
44136fa355b3: Download complete
9104978d9b54: Download complete
Digest: sha256:f296d729ad095d8d5aa3e07eb70d7d40be87ba9c435154c4bbe1d6a888216628
Status: Downloaded newer image for jenkins/jenkins:latest
docker.io/jenkins/jenkins:latest
root@ip-172-31-4-171:~# docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
4f4fb700ef54: Pull complete
5f7e0559360d: Pull complete
cc3656d3a48b: Pull complete
b86e79e12093: Download complete
7728470ff780: Download complete
Digest: sha256:9026624a61cd25542a402a9e7213dd7dbb39724ac9597e331e6b85362558c079
Status: Downloaded newer image for sonarqube:latest
```

and verified them by using "docker images" command

```
root@ip-172-31-4-171:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
jenkins/jenkins:latest	f296d729ad09	707MB	243MB	
nginx:latest	0d4374c710a9	241MB	66.2MB	
sonarqube:latest	9026624a61cd	2.53GB	1.09GB	
tomcat:latest	72400e3f4325	589MB	158MB	
ubuntu:latest	2260313b31c8	160MB	45.3MB	

```
root@ip-172-31-4-171:~#
```

4) Run nginx container and expose on port 81.

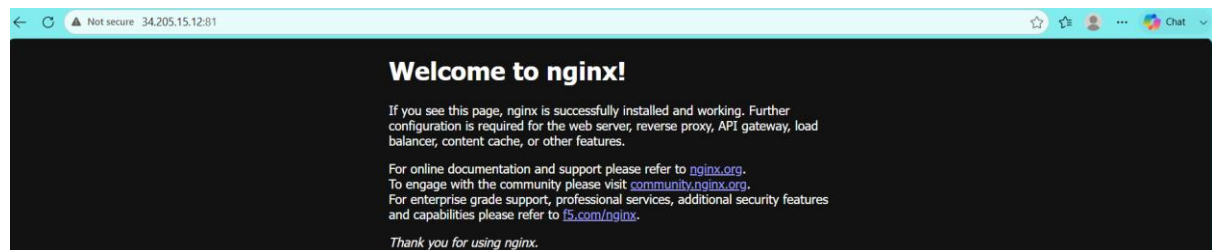
"docker run -d -p 81:80 nginx" → used this command to change port number of nginx from 80 to 81

```
root@ip-172-31-4-171:~# docker run -d -p 81:80 nginx
3eeaf34205be88c580e69b2dbe4aff96471889f0bf4a8770b75768ec90bedd8c
root@ip-172-31-4-171:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
3eeaf34205be	nginx	"/docker-entrypoint..."	9 seconds ago	Up 8 seconds	0.0.0.0:81->80/tcp, [::]:81->80/tcp	jovial_keldysh

```
root@ip-172-31-4-171:~#
```

Verified it on browser by using the public and changed port



5) Delete the apache tomcat image from local.

Checked the pulled images by using → "docker images command"

Deleted tomcat image by using → "docker rmi tomcat command"

Verified it

```
root@ip-172-31-4-171:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
jenkins/jenkins:latest	f296d729ad09	707MB	243MB	
nginx:latest	0d4374c710a9	241MB	66.2MB	
sonarqube:latest	9026624a61cd	2.53GB	1.09GB	
tomcat:latest	72400e3f4325	589MB	158MB	
ubuntu:latest	2260313b31c8	160MB	45.3MB	

```
root@ip-172-31-4-171:~# docker rmi tomcat
Untagged: tomcat:latest
Deleted: sha256:72400e3f4325d05f26a91bacf1ddce2237153ddb8a80304a493e749b6e292a89
root@ip-172-31-4-171:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
jenkins/jenkins:latest	f296d729ad09	707MB	243MB	
nginx:latest	0d4374c710a9	241MB	66.2MB	
sonarqube:latest	9026624a61cd	2.53GB	1.09GB	
ubuntu:latest	2260313b31c8	160MB	45.3MB	

```
root@ip-172-31-4-171:~#
```

6) Inspect the jenkins image, sonarqube image.

To get the full details of an image use inspect command here for Jenkins I have used → "docker inspect jenkins/jenkins" command

```

root@ip-172-31-4-171:~# docker inspect jenkins/jenkins
[
  {
    "Id": "sha256:f296d729ad095d8d5aa3e07eb70d7d40be87ba9c435154c4bbe1d6a888216628",
    "RepoTags": [
      "jenkins/jenkins:latest"
    ],
    "RepoDigests": [
      "jenkins/jenkins@sha256:f296d729ad095d8d5aa3e07eb70d7d40be87ba9c435154c4bbe1d6a888216628"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-08-18T12:18:37.910721617Z",
    "Config": {
      "User": "jenkins",
      "ExposedPorts": {
        "50000/tcp": {},
        "8080/tcp": {}
      },
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",
        "JENKINS_HOME=/var/jenkins_home",
        "JENKINS_SLAVE_AGENT_PORT=50000",
        "REF=/usr/share/jenkins/ref",
        "JENKINS_UC=https://updates.jenkins.io",
        "JENKINS_UC_EXPERIMENTAL=https://updates.jenkins.io/experimental",
        "JENKINS_INCREMENTALS_REPO_MIRROR=https://repo.jenkins-ci.org/incrementals",
        "COPY_REFERENCE_FILE_LOG=/var/jenkins_home/copy_reference_file.log",
        "JAVA_HOME=/opt/java/openjdk",
        "JENKINS_VERSION=2.578"
      ]
    }
  ]
]

```

For sonarqube used → "docker inspect sonarqube" command

```

root@ip-172-31-4-171:~# docker inspect sonarqube
[
  {
    "Id": "sha256:9026624a61cd25542a402a9e7213dd7dbb39724ac9597e331e6b85362558c079",
    "RepoTags": [
      "sonarqube:latest"
    ],
    "RepoDigests": [
      "sonarqube@sha256:9026624a61cd25542a402a9e7213dd7dbb39724ac9597e331e6b85362558c079"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-08-21T18:59:27.786417286Z",
    "Config": {
      "User": "sonarqube",
      "ExposedPorts": {
        "9000/tcp": {}
      },
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "JAVA_HOME=/opt/java/openjdk",
        "LANG=en_US.UTF-8",
        "LANGUAGE=en_US:en",
        "LC_ALL=en_US.UTF-8",
        "JAVA_VERSION=jdk-25.0.4+7",
        "DOCKER_RUNNING=true",
        "SONARQUBE_HOME=/opt/sonarqube",
        "SONAR_VERSION=26.8.0.126808",
        "SQ_DATA_DIR=/opt/sonarqube/data",
        "SQ_EXTENSIONS_DIR=/opt/sonarqube/extensions",
        "SQ_LOGS_DIR=/opt/sonarqube/logs",
      ]
    }
  ]
]

```

7) Run jenkins container and run one sample job.

docker run -d -p 8000:8000 -p 50000:50000 jenkins/Jenkins

docker run → Create and start a container

-d → Run in background (detached mode)

-p 8000:8000 → Map port 8000 (host) → 8000 (container) (Jenkins Web UI)

-p 50000:50000 → Map port 50000 → 50000 (Jenkins agents)

jenkins/jenkins → Official Jenkins Docker image

```

root@ip-172-31-4-171:~# docker run -d -p 8080:8080 -p 50000:50000 jenkins/jenkins
2098b18efce8788d0d38f5696923aad54e25779a9313e045411a2221b5279e
root@ip-172-31-4-171:~# docker ps
CONTAINER ID   IMAGE          NAMES      COMMAND                  CREATED        STATUS        PORTS
2098b18efce8   jenkins/jenkins  "usr/bin/tini -- /u..." 8 seconds ago   Up 7 seconds   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
3eeaf34205be   nginx          "docker-entrypoint..." 24 minutes ago  Up 24 minutes   0.0.0.0:81->80/tcp, [::]:81->80/tcp
root@ip-172-31-4-171:~#

```

Took the public ip and 8080 port and accessed it on browser opened Jenkins console page



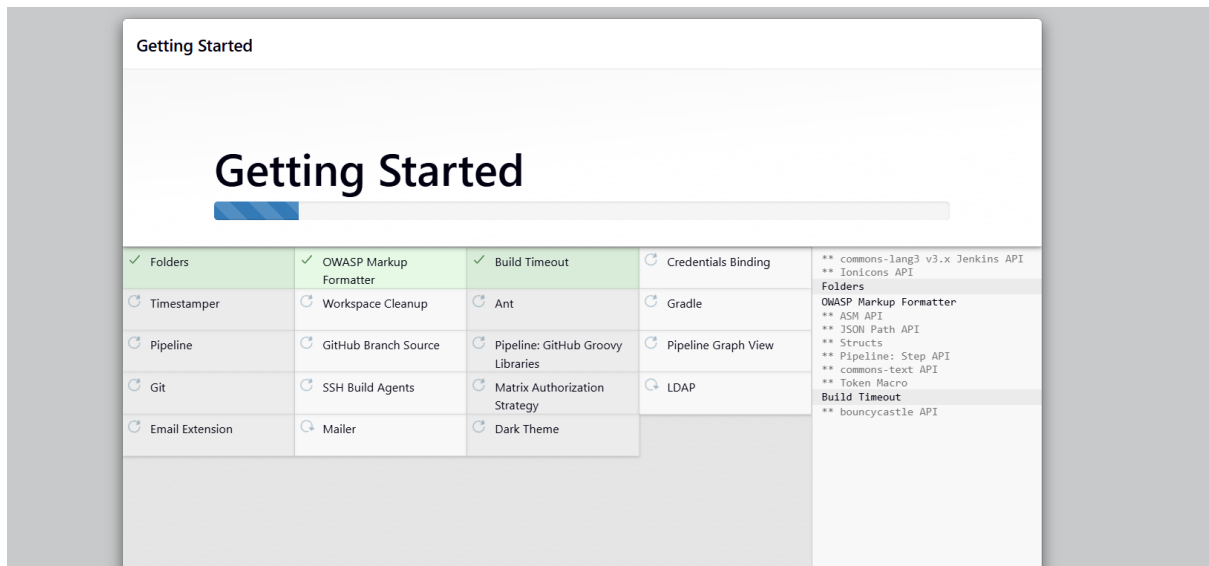
```
docker exec -it busy_booth cat /var/jenkins_home/secrets/initialAdminPassword
```

Part	Meaning
docker exec	Run a command inside a running container
-it	Interactive terminal
busy_booth	Container name (your Jenkins container)
cat	Display file content
/var/jenkins_home/secrets/initialAdminPassword Jenkins password file	

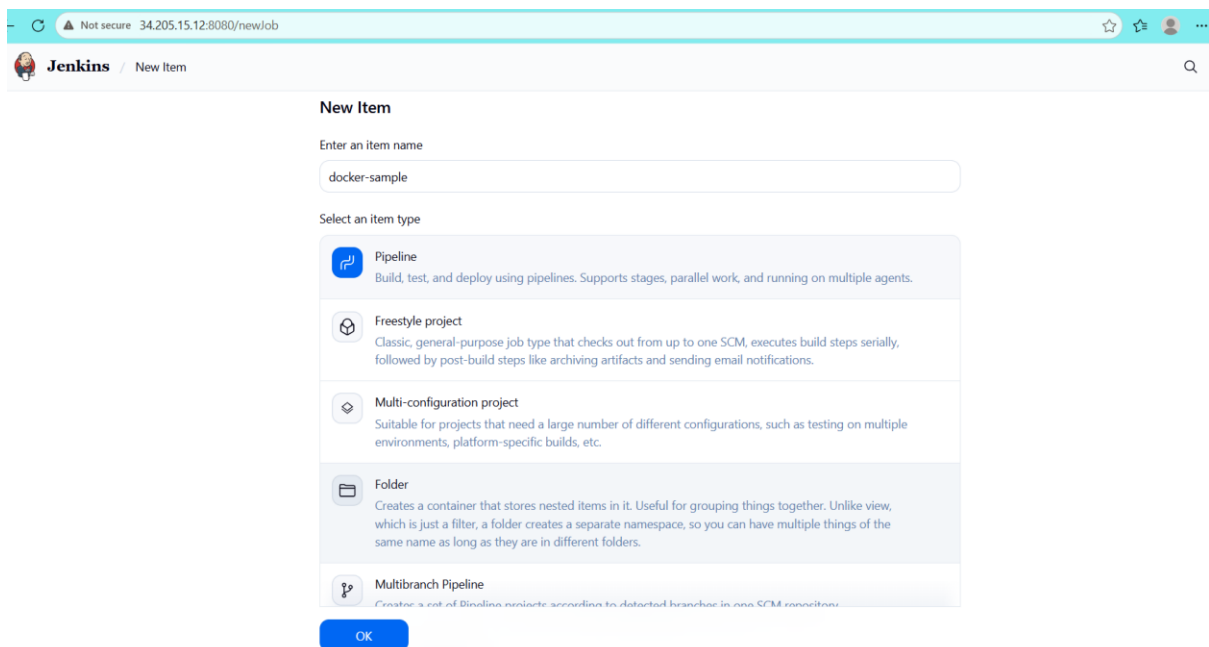
```

root@ip-172-31-4-171:~# docker ps -a
CONTAINER ID   IMAGE          NAMES      COMMAND                  CREATED        STATUS        PORTS
2098b18efce8   jenkins/jenkins  "usr/bin/tini -- /u..." 6 minutes ago   Up 6 minutes   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
e92a9feb0d42   jenkins/jenkins  "usr/bin/tini -- /u..." 12 minutes ago  Created
3eeaf34205be   nginx          "docker-entrypoint..." 31 minutes ago  Up 31 minutes   0.0.0.0:81->80/tcp, [::]:81->80/tcp
root@ip-172-31-4-171:~# docker exec ecstatic_keldysh cat /var/jenkins_home/secrets/initialAdminPassword
8b77775de5c94ed597a4ec66f8e9221b
root@ip-172-31-4-171:~#

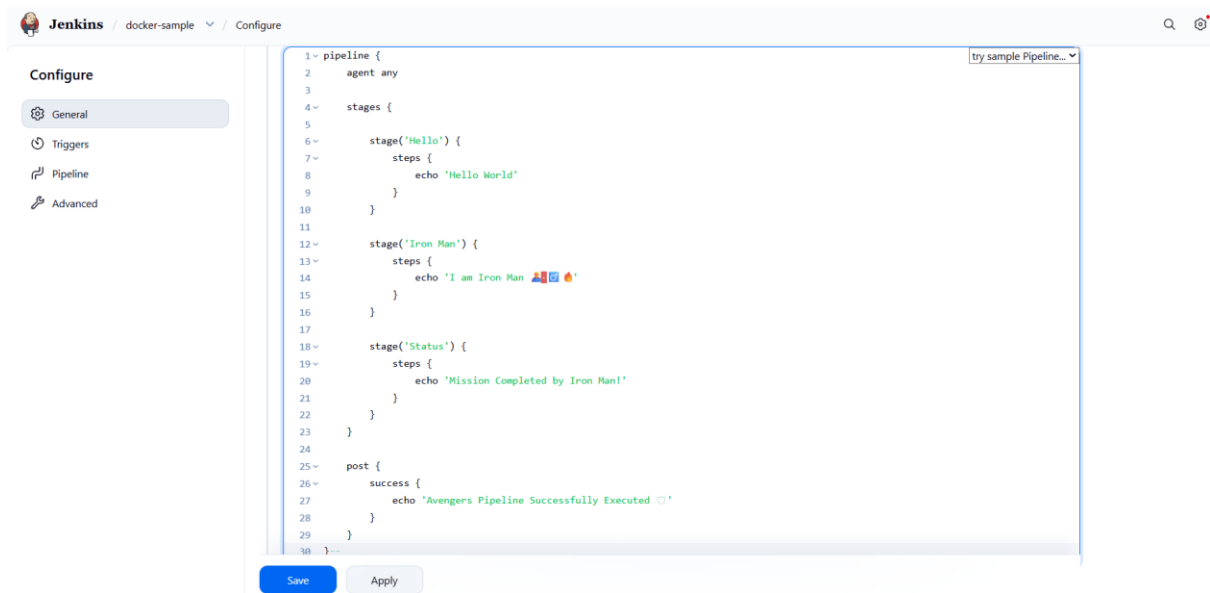
```



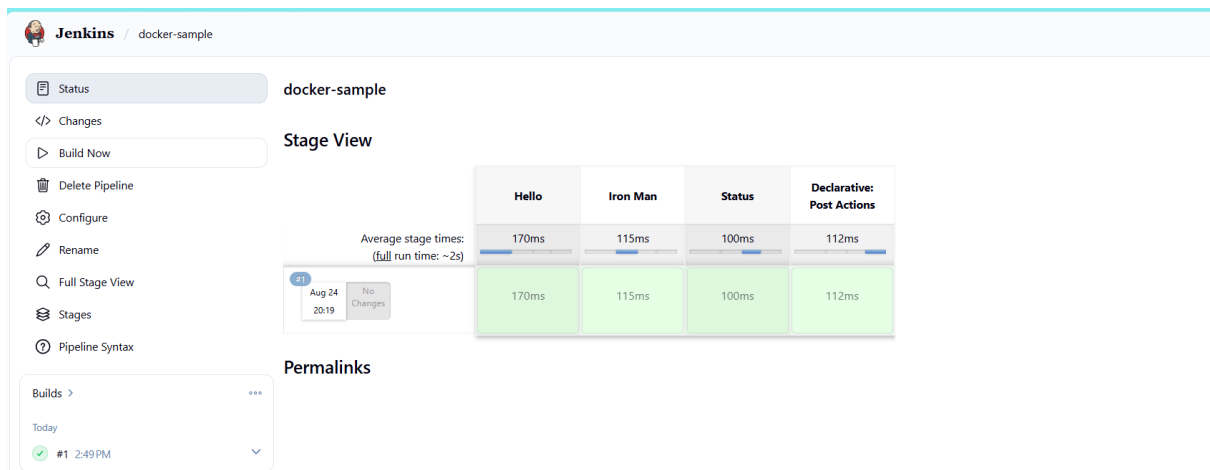
Created a new pipeline job named docker-sample



Used a sample pipeline and in job configuration



The build was success and it was shown in stage view



This was the console output

Jenkins

docker-sample

#1

Status

Changes

Console Output

Delete build #1'

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

Started by user **admin**

[Pipeline] Start of Pipeline

[Pipeline] node

Running on **Jenkins** in /var/jenkins_home/workspace/docker-sample

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Hello)

[Pipeline] echo

Hello World

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Iron Man)

[Pipeline] echo

I am Iron Man 🦾🔥

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Status)

[Pipeline] echo

Mission Completed by Iron Man!

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Declarative: Post Actions)

[Pipeline] echo

Avengers Pipeline Successfully Executed 🦹

[Pipeline] }

[Pipeline] // stage

[Pipeline] }

[Pipeline] // node

[Pipeline] End of Pipeline

Finished: SUCCESS